Madhan Kumar M

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Profile Summary

- Automation Engineer with 3+ years of experience in designing, developing, and maintaining scalable automation frameworks for web and mobile applications.
- Skilled and motivated professional in Automation Testing, specializing in Selenium, Playwright, Java, JavaScript, Appium, Python, and Jenkins.
- Experienced in building Keyword-Driven and Data-Driven frameworks, including Playwright MCP, for complex No-Code / Low-Code platforms.

Experience

Rappit - Automation Test Engineer **May 2025 - Present**

1. Contribution to Quality Assurance:
   - Improved overall test reliability and coverage by implementing automation for complex No-Code / Low-Code platform workflows, reducing manual validation effort by 50%.
2. Testing Framework Implementation:
   - Designed and developed a Keyword-Driven Automation Framework using Playwright, JavaScript, and MCP, enabling different test execution and improving framework scalability by 40%.
   - Implemented Playwright MCP - based automation to handle complex UI interactions, dynamic locators, and application state management, significantly improving test stability and consistency.
   - Developed Keyword-Driven execution using Excel and Google Sheets, allowing all test case execution without code changes and reducing maintenance effort by 35%.
   - Built reusable utilities, custom assertions, and centralized error handling, improving test stability and execution success rate by 30%.
   - Integrated advanced logging, screenshots, video recording, and custom reporting, increasing test analysis efficiency by 25%.
3. Continuous Learning and Adoption of Technologies:
   - Actively adopted advanced Playwright capabilities, framework optimization techniques, and automation best practices to enhance execution speed by 30% and reduce script development time by 25%.
4. Test Cycle Optimization:
   - Optimized regression execution strategy and parallel execution approach, achieving a 40% reduction in overall test cycle time and supporting faster release timelines.

KGiSL Technologies – Automation Test Engineer **December 2022 – May 2025**

1. Contribution to Quality Assurance:
   - Improved testing efficiency by 30% with automated test suites and streamlined processes
2. Testing Framework Implementation*:*
   - Implemented a Selenium testing framework with Java and Cucumber BDD, leading to a 50% reduction in regression testing time across number of projects, ensuring a consistent 95% test coverage.
   - Developed JavaScript frameworks using Playwright for web and WebDriverIO for mobile automation, improving test execution flexibility by 40%.
   - Leveraged a Data-Driven Development (DDD) approach, reducing maintenance time by 30%.
   - Integrated robust reporting, increasing test analysis efficiency by 25%.
3. Continuous Learning and Adoption of Technologies:
   - Continuously updated automation testing skills, learning and adopting tools like Python, Playwright, WebDriverIO, Cucumber, Appium, and UiPath, leading to a 30% increase in test execution efficiency and a 20% reduction in script development time.
4. Achieved a 40% reduction in test cycle time by identifying and resolving key issues, contributing to faster product delivery schedules.
5. Integrated automation scripts into CI/CD pipelines to ensure seamless deployment and validation.

Projects

Alliance Insurance | *Playwright, JavaScript* **January 2023 - Present**

- Currently serving as a QA Test Automation Engineer at KGISL Technologies, contributing to a Alliance Life Insurance project by achieving 95% test coverage through automation.
- Designed and executed over 150 automated test scenarios, ensuring robustness and reliability using Selenium, Playwright, Java, and JavaScript.
- Enhanced the automation framework by incorporating industry best practices, resulting in 30% faster script development and improved scalability.

Awards & Recognitions

- **Champion Award (2024):** Recognized for exceptional contributions at KGISL Technologies by developing a new automation framework, improving test execution efficiency by 30%, reducing manual effort by 40%, and significantly enhancing the scalability of testing processes.

Technical Skills

Technologies/Frameworks: Selenium, Playwright, Appium, UiPath, WebDriver IO, Cucumber, TestNG

Languages: Java, JavaScript, TypeScript, Python, SQL.

Tools: GitHub, Jenkins, Jira, Postman.

Education

Sri Shakthi Institute of Engineering and Technology **August 2018 – May 2022**

Bachelor of Engineering (Computer Science) – CGPA: 7.82 *Coimbatore, Tamil Nadu*

Extracurricular / Certifications

- Published Paper of "Gesture Virtual Mouse Using AI "in GIS Science Journal in April 2022
- Selenium + Cucumber & TestNG Frameworks (Udemy Certifications).
- POSTMAN API Testing (Udemy Certifications).
- ISTQB Foundation Level 2023 Complete Training (Udemy Certifications).